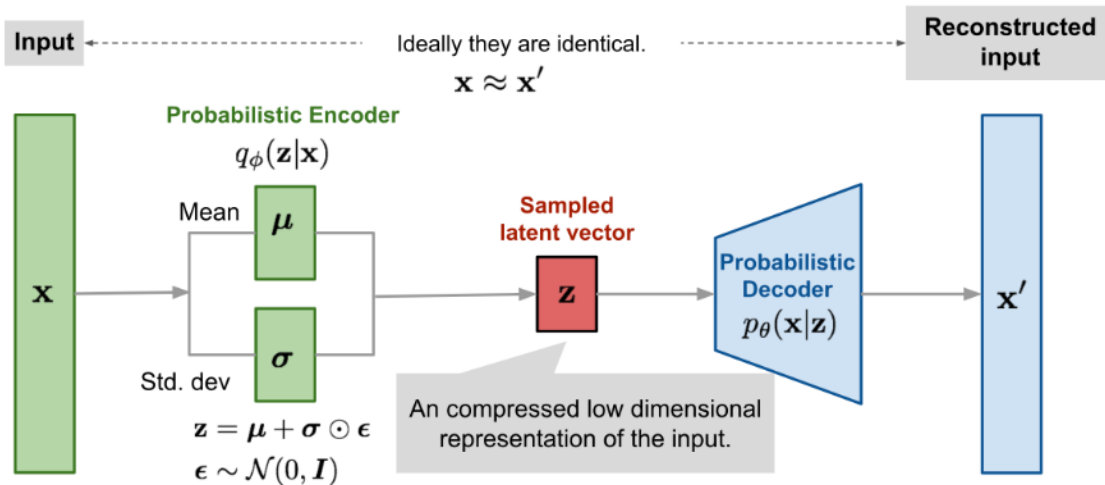


Problem with our loss function: আমাদের loss function ছিল univariate কিন্তু image এর প্রতিটি intensity value হচ্ছে একক variable, তাই আমাদের loss 5^n কে multi-variate এ convert করা সম্ভব।

Today lecture:

1. Intuition of loss function

2. Derivation KL Divergence for multivariate GD



$$L_{VAE}(\theta, \phi) = -\mathbb{E}_{z \sim q_{\phi}(z|x)} \log p_{\theta}(x|z) + D_{KL}(q_{\phi}(z|x) || p_{\theta}(z))$$

we can say $z \sim N(1,0)$
 learn করতে হয় করে $\rightarrow N(1,0)$

To illustrate the effects of using **only reconstruction loss**, **only KL divergence loss**, or **both** in a **Variational Autoencoder (VAE)**, we can consider how each scenario influences the learning process and the quality of the generated outputs.

1. Only Reconstruction Loss [Act like auto encoder]

When training a VAE using only reconstruction loss, the model focuses solely on minimizing the difference between the original input and its reconstruction. This can lead to:

- ✓ **Overfitting:** The model may memorize the training data without learning a meaningful latent representation.
- ✓ **Poor Generalization:** The latent space may not be structured, making it difficult to generate new samples that resemble the training data.
- ✓ **Loss Function:** The loss function would look like this:

Total Loss = Reconstruction Loss

2. Only KL Divergence Loss

Using only KL divergence loss encourages the model to shape the latent space to resemble a standard normal distribution but ignores how well it reconstructs the input data. This leads to:

- **Underfitting:** The model may not learn to represent the input data effectively.
- **Poor Reconstruction Quality:** Outputs may be random or nonsensical since there is no penalty for poor reconstructions.
- **Loss Function:** The loss function would look like this:

Total Loss=KL Divergence

3. Both Reconstruction and KL Divergence Loss

When both losses are combined, the VAE achieves a balance between accurately reconstructing inputs and maintaining a structured latent space. This leads to:

- **Better Generalization:** The model learns useful representations that can generate new data similar to the training set.
- **Controlled Latent Space:** The latent variables are regularized to follow a Gaussian distribution, allowing for smooth interpolation between points in latent space.
- **Loss Function:** The total loss function is formulated as:

Total Loss=Reconstruction Loss+KL Divergence

image generation
কিছু
Normal distribution
বানানোর দিকে মনোযোগ
দিয়ে।

2. Derivation KL of multivariate GD

Univariate Gaussian: Lets we have a single/ univariate random variable X , height of students in a class, then its gaussian distribution can be define using the following formula

General Formulation of Univariate Gaussian Distribution

The probability density function (pdf) of a univariate Gaussian distribution is given by:

$$p(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$X \sim N(0, 1),$$

$$p(x) = \frac{1}{\sqrt{2\pi(1)}} e^{-\frac{1}{2}\left(\frac{x-0}{1}\right)^2}$$

This simplifies to:

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

- Gaussian or normal distribution, 1D

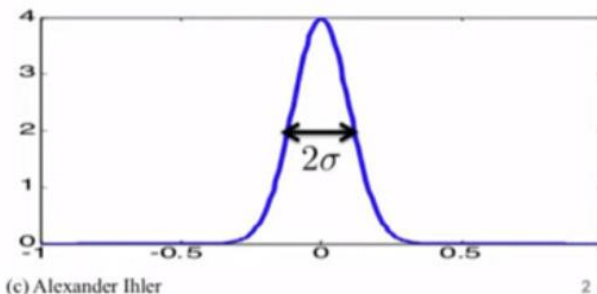
$$\mathcal{N}(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}(x - \mu)^2/\sigma^2\right]$$

- Parameters: mean μ , variance σ^2
(standard deviation σ)

Maximum Likelihood estimates

$$\hat{\mu} = \frac{1}{N} \sum_i x^{(i)}$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_i (x^{(i)} - \hat{\mu})^2$$



↳ (Here, x is random variable)

✓ Multivariate Gaussian:

- Flatten image X is example multivariate random variable that have of multivariate probability function, where every pixel is a random variable that have a sample space of $[0 \text{ to } 255]$

vector of
Random
variables

$$p(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

$d \rightarrow$ (size of image) should be squared

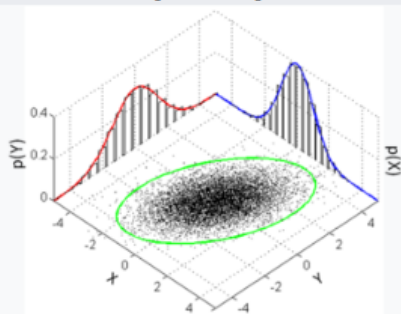
means of those
random variables

Multivariate Gaussian: Let $\mathbf{X} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 & 0.5 \\ 0.5 & 2 \end{bmatrix}\right)$. Joint PDF is: $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

$$p(\mathbf{X}) = \frac{1}{2\pi\sqrt{1.75}} \exp\left(-\frac{1}{2} \begin{bmatrix} X_1 \\ X_2 - 1 \end{bmatrix}^T \begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 - 1 \end{bmatrix}\right)$$

Multivariate normal

Probability density function



Many sample points from a multivariate normal distribution with $\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 3/5 \\ 3/5 & 2 \end{bmatrix}$, shown along with the 3-sigma ellipse, the two marginal distributions, and the two 1-d histograms.

Notation $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

Parameters $\boldsymbol{\mu} \in \mathbb{R}^k$ — location
 $\boldsymbol{\Sigma} \in \mathbb{R}^{k \times k}$ — covariance (positive semi-definite matrix)

Support $\mathbf{x} \in \boldsymbol{\mu} + \text{span}(\boldsymbol{\Sigma}) \subseteq \mathbb{R}^k$

PDF $(2\pi)^{-k/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$,
exists only when $\boldsymbol{\Sigma}$ is positive-definite

Variance vs Covariance matrix:

4. Example for $\mathbf{X} = [2, 4, 6]^T$:

(a) Compute Mean:

$$\bar{x} = \frac{1}{3}(2 + 4 + 6) = 4$$

(b) Subtract Mean:

$$\mathbf{X} - \bar{x} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} - 4 = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}$$

(c) Compute Variance:

$$\text{Var}(X) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{3-1} [(-2)^2 + 0^2 + 2^2]$$

$$\text{Var}(X) = \frac{1}{2}(4 + 0 + 4) = \frac{8}{2} = 4$$

(d) Covariance Matrix (Single Variable Case):

The covariance matrix is just a 1×1 scalar:

$$\Sigma = [4]$$

5. For Multivariable Data Example:

If you had data $\mathbf{X} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{bmatrix}$, representing two variables:

(a) Compute Column-wise Means:

$$\bar{\mathbf{X}} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \quad [4 \ 3]$$

(b) Subtract Mean:

$$\mathbf{X} - \bar{\mathbf{X}} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \\ 6 & 5 \end{bmatrix} - \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 0 & 0 \\ 2 & 2 \end{bmatrix}$$

Handwritten notes: [4 3] above the second matrix, 2-4 and 1-3 above the first two rows of the result matrix.

(c) Compute Covariance Matrix:

$$\Sigma = \frac{1}{n-1} (\mathbf{X} - \bar{\mathbf{X}})^T (\mathbf{X} - \bar{\mathbf{X}})$$

Handwritten note: x^2 (with an arrow pointing to the exponent) and $(x^T x \text{ in matrix})$ next to the formula.

For this example:

$$\Sigma = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

This covariance matrix indicates high correlation between the two variables.

✓ Derivation of KL divergence between the multivariate Gaussian distribution:

We want to calculate the KL divergence between two distributions:

$$p(x) \sim \mathcal{N}(\mu_1, \Sigma_1), \quad q(x) \sim \mathcal{N}(\mu_2, \Sigma_2).$$

The formula for KL divergence is:

$$D_{KL}(p(x) \| q(x)) = \int p(x) \log \frac{p(x)}{q(x)} dx.$$

Integral means
we have
continuous
Random variable

✓ Step 1: Write down the Gaussian PDFs

The probability density function (PDF) of a multivariate Gaussian distribution is:

$$p(x) = \frac{1}{(2\pi)^{k/2} |\Sigma_1|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) \right),$$

$$q(x) = \frac{1}{(2\pi)^{k/2} |\Sigma_2|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) \right),$$

where:

- k is the dimensionality of x ,
- $|\Sigma_1|$ and $|\Sigma_2|$ are the determinants of the covariance matrices,

✓ $(x - \mu)^T \Sigma^{-1} (x - \mu)$ is the Mahalanobis distance.

✓ The Mahalanobis Distance is a measure of distance between a point and a distribution. Unlike the Euclidean distance, it accounts for the correlations and variances of the data dimensions, making it especially useful for multivariate data.

Step 2. Compute the ratio $\frac{p(x)}{q(x)}$ (KL-Divergence formula)

The ratio of $p(x)$ and $q(x)$ is:

$$\frac{p(x)}{q(x)} = \frac{|\Sigma_2|^{1/2}}{|\Sigma_1|^{1/2}} \exp \left(-\frac{1}{2}(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) + \frac{1}{2}(x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) \right).$$

Taking the logarithm of the ratio:

$$\log \frac{p(x)}{q(x)} = \frac{1}{2} \log \frac{|\Sigma_2|}{|\Sigma_1|} - \frac{1}{2}(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) + \frac{1}{2}(x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2).$$

Step 3. Substitute into the KL divergence formula

Now substitute the expression for $\log \frac{p(x)}{q(x)}$ into the KL divergence formula:

$$D_{KL}(p(x) \| q(x)) = \int p(x) \left[\frac{1}{2} \log \frac{|\Sigma_2|}{|\Sigma_1|} - \frac{1}{2}(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) + \frac{1}{2}(x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) \right] dx.$$

Separate the terms of the integral into three parts:

$$D_{KL}(p(x) \| q(x)) = \frac{1}{2} \log \frac{|\Sigma_2|}{|\Sigma_1|} + I_1 - I_2, \quad (p(x) \text{ over } \Sigma_1 \text{ and } \Sigma_2)$$

where:

$$I_1 = \int p(x) \left[\frac{1}{2}(x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) \right] dx,$$

$$I_2 = \int p(x) \left[\frac{1}{2}(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) \right] dx.$$

$$\int p(x) \frac{1}{2} \log \frac{|\Sigma_2|}{|\Sigma_1|} dx$$

Step 4: Evaluate each term

(a) First Term: $\log \frac{|\Sigma_2|}{|\Sigma_1|}$

This term is constant and independent of x , so it directly contributes:

$$\frac{1}{2} \log \frac{|\Sigma_2|}{|\Sigma_1|}$$

$$= \frac{1}{2} \log \frac{|\Sigma_2|}{|\Sigma_1|} \int p(x) dx$$

(b) Second Term: $\int p(x) (x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) dx$

Using the expectation property of the Gaussian distribution:

$$\mathbb{E}_{p(x)}[(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1)] = \text{tr}(\Sigma_1^{-1} \Sigma_1) = k,$$

where k is the dimensionality of the space.

Thus, the integral:

$$I_2 = \frac{1}{2} k.$$

$$\mathbb{E}_x[f(x)] = \int p(x) f(x) dx$$

Relation between trace of a matrix vs expectation

$$\text{tr} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1+1=2$$

$$\text{tr} [2] = 2$$

If, $x = \text{constant}$ then,

$$\mathbb{E}[x] = x \text{ or } \mathbb{E}[x] = \mathbb{E}[\text{tr}(x)]$$

$$\text{Property of trace: } \text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA) \neq \text{tr}(ACB)$$

$$\text{cyclic rotate possible}$$

$$\mathbb{E}[(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1)] \text{ constant}$$

$$\mathbb{E}[\text{tr}[(x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1)]] \text{ (constant property of expectation)}$$

$$\mathbb{E}[\text{tr}(\Sigma_1^{-1} (x - \mu_1) (x - \mu_1)^T)] \text{ (Rotate property of trace)}$$

$$\mathbb{E}[\text{tr}(\Sigma_1^{-1} \Sigma_1)]$$

$$= k \times k$$

$$[]_{k \times k}$$

identity matrix

sum = k

$\int p(x) dx$ because, $p(x)$ is a probability distribution so, because,

$\int p(x) dx$ value is 1, because, probability sum is 1. From property of probability.

(Equation 01)

$\frac{1}{A} \frac{1}{B} \mu_1 - \mu_2$
 $(x - \frac{\mu_1}{A} + \frac{\mu_1 - \mu_2}{B})^T \Sigma_2^{-1} (\frac{x - \mu_1}{A} + \frac{\mu_1 - \mu_2}{B})$
 $(x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) + \dots + \dots$

(c) Third Term: $\int p(x) (x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) dx$

Expand $(x - \mu_2)$ as $(x - \mu_1) + (\mu_1 - \mu_2)$:

$$(x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) = \frac{(x - \mu_1)^T \Sigma_2^{-1} (x - \mu_1)}{1} + \frac{2(x - \mu_1)^T \Sigma_2^{-1} (\mu_1 - \mu_2)}{2} + \frac{(\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2)}{1}$$

Take the expectation under $p(x)$:

- $\mathbb{E}_{p(x)}[(x - \mu_1)^T \Sigma_2^{-1} (x - \mu_1)] = \text{tr}(\Sigma_2^{-1} \Sigma_1)$
- $\mathbb{E}_{p(x)}[2(x - \mu_1)^T \Sigma_2^{-1} (\mu_1 - \mu_2)] = 0$ (since $\mathbb{E}_{p(x)}[x - \mu_1] = 0$, **cause $E_X[x] = \mu_2$**)
- $\mathbb{E}_{p(x)}[(\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2)] = (\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2)$

সবগুলো আলাদা করে $p(x)$ করে

see lecture 2

Thus:

$$I_1 = \frac{1}{2} [\text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2)]$$

Step 5: Combine All Terms

(KL Divergence of Multivariate Gaussian Distribution)

Finally, substitute everything back:

$$\underline{D_{KL}(p(x) \| q(x))} = \frac{1}{2} \left[\log \frac{|\Sigma_2|}{|\Sigma_1|} - k + \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_1 - \mu_2)^T \Sigma_2^{-1} (\mu_1 - \mu_2) \right]$$

This is the final result.

For KL Divergence for VAE:

To compute $D_{KL}[q(z|x)||p(z)]$ for the VAE case using the given formulation:

$$D_{KL}(p(x)||q(x)) = \frac{1}{2} \left[\log \frac{|\Sigma_2|}{|\Sigma_1|} - k + \text{tr}(\Sigma_2^{-1}\Sigma_1) + (\mu_1 - \mu_2)^T \Sigma_2^{-1}(\mu_1 - \mu_2) \right],$$

where:

- ✓ $\mu_1 = \mu$ and $\Sigma_1 = \Sigma$ come from $q(z|x)$,
- ✓ $\mu_2 = 0$ and $\Sigma_2 = I$ come from $p(z)$. $\rightarrow N(0, I) \cdot p(z)$

Step-by-Step Calculation

1. Log Determinant Term: $\log \frac{|\Sigma_2|}{|\Sigma_1|}$

- $\Sigma_2 = I$, so $|\Sigma_2| = 1$,
- $|\Sigma_1| = |\Sigma|$,

$$\checkmark \left[\log \frac{|\Sigma_2|}{|\Sigma_1|} = -\log |\Sigma| \right]$$

2. Trace Term: $\text{tr}(\Sigma_2^{-1}\Sigma_1)$

- $\Sigma_2 = I$, so $\Sigma_2^{-1} = I$,
- Therefore, $\text{tr}(\Sigma_2^{-1}\Sigma_1) = \text{tr}(\Sigma_1) = \text{tr}(\Sigma)$.

3. Quadratic Term: $(\mu_1 - \mu_2)^T \Sigma_2^{-1}(\mu_1 - \mu_2)$

- $\mu_1 = \mu$, $\mu_2 = 0$, and $\Sigma_2^{-1} = I$,
- So, $(\mu_1 - \mu_2)^T \Sigma_2^{-1}(\mu_1 - \mu_2) = \mu^T I \mu = \mu^T \mu$.

4. Combine Terms

Plugging all these into the KL divergence formula:

$$D_{KL}[q(z|x)||p(z)] = \frac{1}{2} [-\log |\Sigma| - k + \text{tr}(\Sigma) + \mu^T \mu] .$$

✓ Final Result:

$$D_{KL}[q(z|x)||p(z)] = \frac{1}{2} [-\log |\Sigma| - k + \text{tr}(\Sigma) + \|\mu\|^2] .$$

This is the KL divergence between the approximate posterior $q(z|x)$ and the prior $p(z)$ in a VAE.

Now the final loss function of VAE:

Loss Function with the Computed KL Divergence

The VAE loss is:

$$\mathcal{L}_{\text{VAE}} = -\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] + D_{KL}[q_{\phi}(z|x)||p(z)].$$

For D_{KL} , using the formula for two Gaussian distributions $q(z|x) = \mathcal{N}(\mu, \Sigma)$ and $p(z) = \mathcal{N}(0, I)$, we computed:

$$D_{KL}[q_{\phi}(z|x)||p(z)] = \frac{1}{2} (-\log |\Sigma| - k + \text{tr}(\Sigma) + \|\mu\|^2).$$

Incorporating into the Loss

Substituting the KL divergence into the VAE loss:

$$\mathcal{L}_{\text{VAE}} = -\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] + \frac{1}{2} (-\log |\Sigma| - k + \text{tr}(\Sigma) + \|\mu\|^2).$$

1. Reconstruction Loss ($-\mathbb{E}_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)]$):

- For Gaussian decoders, this term can often be approximated by the Mean Squared Error (MSE):

$$\text{Reconstruction Loss} \approx \frac{1}{2} \|x - \hat{x}\|^2,$$

where $\hat{x}$ is the reconstructed input.

2. KL Divergence:

- From the computation:

$$D_{KL}[q_{\phi}(z|x)||p(z)] = \frac{1}{2} (-\log |\Sigma| - k + \text{tr}(\Sigma) + \|\mu\|^2).$$

Final Loss Function

The final loss function becomes:

$$\mathcal{L}_{\text{VAE}} = \frac{1}{2} \|x - \hat{x}\|^2 + \frac{1}{2} (-\log |\Sigma| - k + \text{tr}(\Sigma) + \|\mu\|^2).$$

Simplifications for Diagonal Covariance Matrix

If Σ is diagonal with entries $\sigma_1^2, \sigma_2^2, \dots, \sigma_k^2$:

1. $\log |\Sigma| = \sum_{i=1}^k \log \sigma_i^2$,
2. $\text{tr}(\Sigma) = \sum_{i=1}^k \sigma_i^2$,
3. The KL divergence simplifies to:

$$D_{KL} = \frac{1}{2} \sum_{i=1}^k (-\log \sigma_i^2 - 1 + \sigma_i^2 + \mu_i^2) .$$

Thus, the loss becomes:

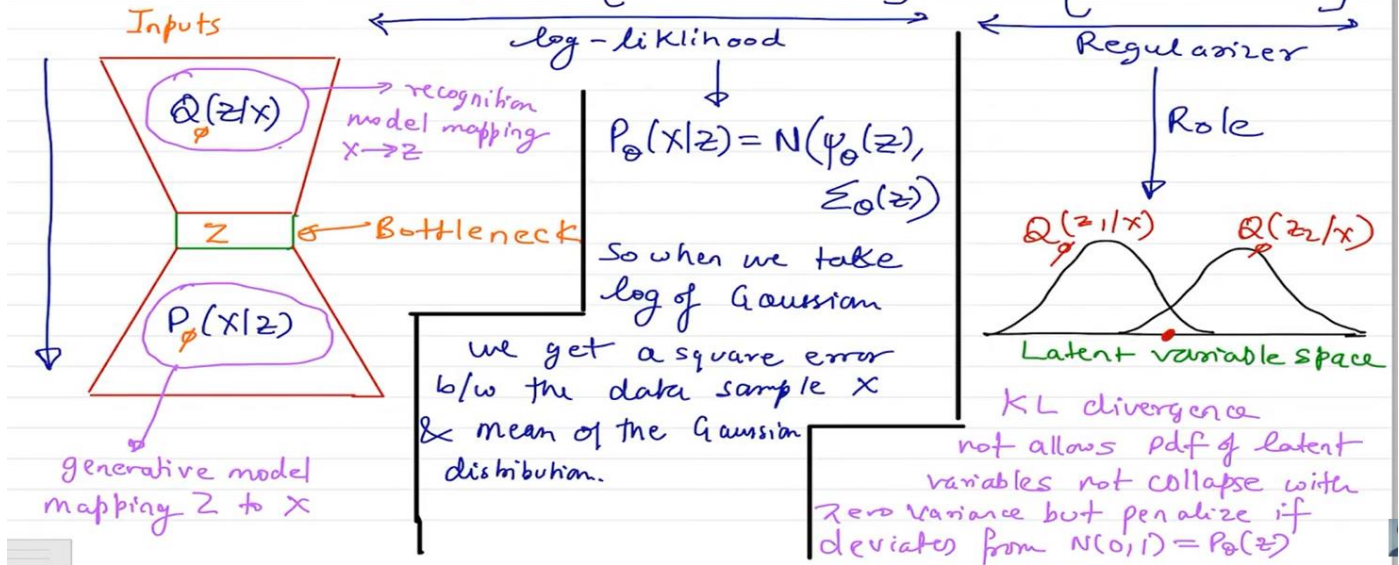
$$\mathcal{L}_{\text{VAE}} = \frac{1}{2} \|x - \hat{x}\|^2 + \frac{1}{2} \sum_{i=1}^k (-\log \sigma_i^2 - 1 + \sigma_i^2 + \mu_i^2) .$$

Interpretation

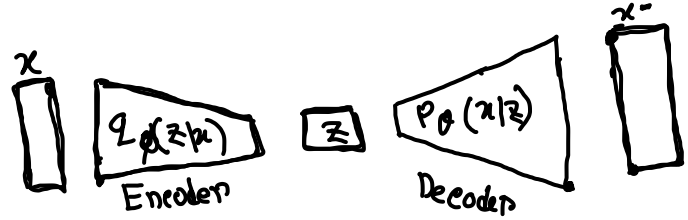
1. **First term:** Ensures $\hat{x}$ (reconstructed input) is close to x (original input).
2. **Second term:** Regularizes the encoder by forcing $q_\phi(z|x)$ to align with the prior $p(z)$.

MORE INTUITION ABOUT LOSS FUNCTION

$$L(\theta, \phi) = -E_{z \sim Q_\phi(z/x)} \left[\log \left(P_\theta(x/z) \right) \right] + D_{KL} \left[Q_\phi(z/x) \parallel P_\theta(z) \right]$$



https://www.youtube.com/watch?v=w8F7_rQZxXk&list=PLdxQ7SoCLQANizknbliHzL_hYjEal-wUe



1. Expression for $P_\theta(X|Z)$:

The likelihood $P_\theta(X|Z)$ is modeled as a Gaussian distribution:

$$P_\theta(X|Z) = \frac{1}{\sqrt{(2\pi)^k |\Sigma_\theta(Z)|}} \exp \left(-\frac{1}{2} (X - \mu_\theta(Z))^T \Sigma_\theta(Z)^{-1} (X - \mu_\theta(Z)) \right)$$

- ✓ $\mu_\theta(Z)$: Mean of the Gaussian, predicted by the decoder network.
- ✓ $\Sigma_\theta(Z)$: Covariance matrix of the Gaussian, often simplified to a diagonal matrix for computational efficiency.
- ✓ X : Input data.
- ✓ Z : Latent variable sampled from the encoder's posterior distribution.

2. Log of the Likelihood:

Taking the logarithm of $P_\theta(X|Z)$ simplifies the expression:

$$\log P_\theta(X|Z) \propto -\frac{1}{2} \underbrace{(X - \mu_\theta(Z))^T}_{\text{row vector}} \underbrace{\Sigma_\theta(Z)^{-1}}_{\text{matrix}} \underbrace{(X - \mu_\theta(Z))}_{\text{column vector}}$$

This step eliminates constant terms (like $\sqrt{(2\pi)^k}$) since they do not affect optimization.

3. Connection to Reconstruction Error:

The term $(X - \mu_\theta(Z))^T \Sigma_\theta(Z)^{-1} (X - \mu_\theta(Z))$ represents the Mahalanobis distance between X and the predicted mean $\mu_\theta(Z)$, weighted by the inverse covariance matrix.

- If $\Sigma_\theta(Z)$ is the identity matrix or a scalar, this simplifies to a squared reconstruction error:

$$\underline{(X - \mu_\theta(Z))^T (X - \mu_\theta(Z))}$$

This measures how well the decoder reconstructs the input X from the latent representation Z .

→ (আমরা, loss function forward pass দিয়ে next ৩ অংশে
backward pass দিয়ে with Reparameterization trick) ←